Committee: Disarmament and International Security Committee (DISEC)

Topic: Regulating the Development of Emerging Technologies that Increase the Risk of Nuclear Escalation

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1. Introduction

The world as we know it is entering a period where technological innovations and developments are transforming the nature of international security. Artificial intelligence, cyber security/attacks, autonomous weapons, hypersonic weapons, and advanced AI-powered surveillance systems can strengthen national security, but they may also create new risks that could contribute to nuclear escalation.

For DISEC, the central challenge is to determine how the international community can manage these risks whilst preserving legitimate technological development and national security.

2. Definition of Essential Terms

a) Nuclear escalation - a process in which a conflict increases in intensity and moves closer to the use of nuclear weapons, either deliberately or through miscalculation or miscommunication.

b) Nuclear Command, Control and Communications (NC3) - the systems/technologies and procedures used to detect threats, communicate orders, and maintain control over nuclear forces.

c) Early-Warning Systems - a network of satellites, radars, sensors, sirens, and many other technologies that are used to detect potential military threats and missile launches.

d) Dual-Use Technology - a technology that can be utilized for both civilian and military purposes, making regulation more complex.

e) Deterrence - a strategy of preventing an opponent from taking hostile actions by maintaining a credible threat of consequences.

3. General Context

For decades, nuclear stability has depended not only on nuclear weapons themselves, but on the systems surrounding them: early-warning networks, communication infrastructure, intelligence systems, command structures, and decision-making procedures. Emerging technologies are now transforming many of these systems.

Artificial intelligence can process large amounts of information and potentially assist in identifying threats. Advanced sensors and satellites can improve early warning capabilities, while cyber technologies can both strengthen and disrupt military infrastructure. Hypersonic weapons can also reduce the time available for states to detect and respond to potential threats. These new technological innovations create a difficult security dilemma.

The Issue of Speed:

In a nuclear crisis, time is critical if new technologies make weapons and decision-making systems significantly faster. During a crisis, political and military leaders must determine whether a perceived threat is genuine, assess the intentions of other states, communicate with relevant authorities and weapons, and determine an appropriate response. Each of these processes requires reliable information and sufficient time for verification. Verification is crucial because acting on incomplete or incorrect information could turn a misunderstanding into a larger conflict. Emerging technologies can significantly increase the speed of weapons, communications, surveillance, and more; the amount of time for human judgment (which is crucial) may decrease. Although greater speed can strengthen a state's ability to respond to genuine threats, it can also create pressure to make decisions before information has been sufficiently verified.

Therefore, the challenge is not simply whether faster technologies improve military responsiveness. States must also consider whether increased speed leaves enough time for communication, verification and human judgement. Maintaining sufficient time for human decision-making is important for reducing the risk of unintended nuclear escalation

The Issue of Uncertainty:

Artificial Intelligence systems can analyse information at never seen before speeds; however, their conclusions depend on the quality and accuracy of the information they receive. In a crisis, information could theoretically be incomplete, biased, contradictory, or affected by deliberate enemy deception; an AI that produces an incorrect assessment could therefore potentially influence human decision-makers during critical moments where it matters most.

The difficulty we face is further enhanced through the complexity of modern military systems. Military and political leaders (the decision-makers) may not always be able to completely comprehend why an AI system has reached a particular conclusion, particularly when systems are processing large amounts of information at the same time. If governments begin to rely heavily on automated assessments, there is a risk that confidence in technological systems could exceed their actual reliability. This creates a fundamental question about how much authority humans should delegate to machines when an error could have severe consequences.

The issue is how AI can be used while remaining subject to appropriate human oversight and verification.

The Cyber aspect:

Nuclear-related infrastructure increasingly depends on digital systems, communication networks, early warning systems, and other technologies. While these systems can improve the speed and efficiency of nuclear command and control, they can also create new vulnerabilities. Cyber operations could potentially disrupt or manipulate important information, which could affect decision-making during a nuclear crisis. This makes cybersecurity an important part of preventing accidental or unintended escalation.

4. Key Parties Involved and their Perspectives

a) United States:

The United States places importance on maintaining nuclear deterrence while protecting nuclear command, control, and communications from cyber and technological threats. It also aims to continue developing AI and other emerging technologies while managing the risks they may create for strategic stability.

b) Russian Federation

The Russian Federation places strong importance on strategic stability, nuclear deterrence, and maintaining control over its military technologies. It is also concerned with protecting nuclear command and control systems from cyber threats and maintaining its strategic capabilities as other countries develop emerging technologies.

c) China

China is rapidly developing capabilities in artificial intelligence, cyber technologies, space systems, and other advanced technologies. It has an interest in continuing technological development while preventing emerging technologies from creating instability or weakening its strategic position.

d) The United Kingdom and France

The United Kingdom and France are nuclear weapons states focused on credible deterrence and risk reduction; both would be relevant regarding discussions on AI, cybersecurity, and strategic stability. They will also most likely support measures addressing accidental or unintended escalation.

e) Pakistan

Pakistan is a nuclear-weapon state with significant interests in maintaining strategic deterrence and national security; its position is particularly shaped by regional security dynamics and the development of advanced military technologies. Pakistan may therefore emphasize strategic stability, risk reduction, and preventing emerging technologies from increasing regional tensions.

f) India

India is a nuclear-armed state with growing capabilities in AI, cyber technologies, surveillance, and advanced military tech/systems. Its interests include retaining regional/strategic defence, deterrence, strengthening national security, and growing and innovating its military systems.

India also has an interest in ensuring that new technologies do not increase regional instability or reduce its strategic autonomy.

5. Questions to Consider

1. What degree of direct human control should be required when artificial intelligence or AI-powered (autonomous) weapons are integrated into nuclear command, control, and communication?

2. Which emerging technologies pose the most significant threat/risks to global/nuclear stability, and should their development be subject to specific international safeguards?

3. How can member states reduce the risk of miscalculation when previously mentioned emerging technologies drastically reduce the time they have for crucial decision-making?

4. What measures should our committee enact to safeguard nuclear command, control and communications and early warning systems from cyberattacks

5. How can member states increase transparency and trust-building between fellow states without necessarily requiring the disclosure of sensitive national security information?

6. What mechanisms can ensure accountability when artificial intelligence contributes to a dangerous and potentially cataclysmic military command miscalculation or escalation?

7. How can our committee prevent competition in emerging military technologies from contributing to a dangerous arms race whilst preserving states' rights to technological advancement/independent development?

8. Should the international community strengthen existing frameworks or establish new international norms specifically addressing the relationship between emerging technologies and nuclear escalation?

6. Suggested Research Resources

a) United Nations General Assembly Resolution A/RES/80/23
https://digitallibrary.un.org/record/4095461?utm_
A key primary source directly addressing the topic. The resolution discusses human control and oversight, shortened decision-making windows, technical vulnerabilities, and potential risks associated with integrating AI into nuclear-related systems

b) Stockholm International Peace Research Institute (SIPRI)
Impact of military artificial intelligence on nuclear escalation risk
https://www.sipri.org/publications/2025/sipri-insights-peace-and-security/impact-military-artificial-intelligence-nuclear-escalation-risk

c) United Nations General Assembly Resolution A/RES/80/57
https://digitallibrary.un.org/record/4095989?ln=en&v=pdf&utm_=
This resolution provides a useful starting point for researching autonomous weapons, human control, accountability, and international approaches to emerging military technologies.

d) United Nations Institute for Disarmament Research (UNIDR)
UNIDIR's research covers cybersecurity, artificial intelligence, emerging technologies, autonomous weapons, disarmament, and international security. Its work is particularly valuable for examining international governance, risk reduction, and technological developments.
https://unidir.org/work/science-tech-innovation/?utm_
https://unidir.org/publication/securing-cyberspace-for-peace-insights-into-cyberthreats-and-international-security-in-2025/?utm_

e) International Institute for Strategic Studies (IISS)

IISS provides research on nuclear deterrence, hypersonic weapons, missile systems, strategic stability, and military technology.

https://www.iiss.org/publications/strategic-comments/2020/hypersonic-weapons-and-strategic-stability/?utm_

7. Conclusion

Emerging technologies are rapidly changing our international security environment. AI, cyber capabilities/security, autonomous systems and weapons, hypersonic weapons, and advanced surveillance systems undoubtedly have the ability to strengthen national security; however, they create issues of uncertainty, shorten decision-making times, and create several new pathways to nuclear escalation.

DISEC's challenge is to reduce these risks without unnecessarily restricting genuine technological development or national security. Moving forward, delegates must consider human oversight, accountability, transparency (without undermining national security), international cooperation, and global strategic stability, all this whilst considering the continuously changing and adapting security interests of states.

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